

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CASE STUDY

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	20% of CIA	Total Marks:	100
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	CH.POORNA TEJA	Reg. No.:	192472149
Section / Batch:		Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given case study questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues, provide an in-depth analysis, and propose innovative, feasible solutions with strong justification.

Question Type	Focus Area	Questions
Case Study	Focus on Design Divide and Conquer algorithms: Merge Sort, Quick Sort, Binary Search and Matrix Multiplication	Q1

SECTION A – CASE STUDY

A scientific application performs repeated matrix multiplications for simulations. Analyze how divide-and-conquer techniques can improve computational efficiency. Identify challenges in handling large matrices. Apply optimized matrix multiplication, such as Strassen's method. Analyze time complexity and compare it with the traditional method. Design a solution and justify its advantages.

Code of Conduct

I P. Sai Charan Tej(192465048)__(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Case Study					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25%	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20%	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20%	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10%	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%				
Application of Concepts	25%				
Analysis	20%				
Solution Design	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Strassen's Matrix Multiplication

- 1. Start**
- 2. Read the order n of the matrices.**
- 3. Read matrix A and matrix B.**
- 4. If the matrix size is 1×1 , multiply the two elements directly.**
- 5. Divide both matrices into four equal submatrices:**
 - **A11, A12, A21, A22**
 - **B11, B12, B21, B22**
- 6. Calculate the seven Strassen products:**
 - **$M1 = (A11 + A22)(B11 + B22)$**
 - **$M2 = (A21 + A22)B11$**
 - **$M3 = A11(B12 - B22)$**
 - **$M4 = A22(B21 - B11)$**
 - **$M5 = (A11 + A12)B22$**
 - **$M6 = (A21 - A11)(B11 + B12)$**
 - **$M7 = (A12 - A22)(B21 + B22)$**
- 7. Calculate the resulting submatrices:**
 - **$C11 = M1 + M4 - M5 + M7$**
 - **$C12 = M3 + M5$**
 - **$C21 = M2 + M4$**
 - **$C22 = M1 - M2 + M3 + M6$**
- 8. Combine C11, C12, C21, C22 to form the final matrix C.**
- 9. Display the resulting matrix.**
- 10. Stop**

Output:-

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
DAA PROGRAMMING

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CO3 AT2-CASE STUDY

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Language: Python C C++ Java

QUESTIONS

Q1

Scientific Computing (Matrix Multiplication) A scientific application performs repeated matrix multiplications for simulations

100 marks

1/1 answered

Analyze how divide and conquer techniques can improve computational efficiency. Identify challenges in handling large matrices. Apply optimized matrix multiplication (such as Strassen's method). Analyze time complexity and compare with the traditional method. Design a solution and justify its advantages.

Your Code

```
1 def add(a, m):
2     n = len(a)
3     return [[a[i][j] + m[i][j] for j in range(n)] for i in range(n)]
4
5
6 def subtract(a, m):
7     n = len(a)
8     return [[a[i][j] - m[i][j] for j in range(n)] for i in range(n)]
9
10
11 def strassen(a, m):
12     n = len(a)
13
14     if n == 1:
15         return [[a[0][0] + m[0][0]]]
16
17     mid = n // 2
18
19     a11 = [row[mid] for row in a[:mid]]
20     a12 = [row[mid] for row in a[mid:]]
21     a21 = [row[mid] for row in a[:mid]]
22     a22 = [row[mid] for row in a[mid:]]
23
24     m11 = [row[mid] for row in a[:mid]]
25     m12 = [row[mid] for row in a[mid:]]
26     m21 = [row[mid] for row in a[:mid]]
27     m22 = [row[mid] for row in a[mid:]]
28
29     m1 = strassen(add(a11, a22), add(m11, m22))
30     m2 = strassen(add(a11, a22), m11)
31     m3 = strassen(m11, subtract(m12, m22))
32     m4 = strassen(a22, subtract(m21, m11))
33     m5 = strassen(add(a11, a12), m22)
34     m6 = strassen(subtract(a21, a11), add(m11, m12))
35     m7 = strassen(subtract(a12, a22), add(m11, m22))
36
37     C11 = add(subtract(add(m1, m4), m5), m7)
38     C12 = add(m4, m5)
39     C21 = add(m6, m4)
40     C22 = add(subtract(add(m1, m3), m6), m6)
41
42     C = []
43
44     for i in range(mid):
45         C.append(C11[i] + C12[i])
46
47     for i in range(mid):
48         C.append(C21[i] + C22[i])
49
50     return C
51
52
53 n = int(input())
54
55 A = []
56 B = []
57
58 for i in range(n):
59     A.append(list(map(int, input().split())))
60
61 for i in range(n):
62     B.append(list(map(int, input().split())))
63
64 result = strassen(A, B)
65
66 for row in result:
67     print(*row)
```

Input

```
2
1 2
3 4
5 6
```

Output

```
19 22
43 30
```

Submitted